

# Part A

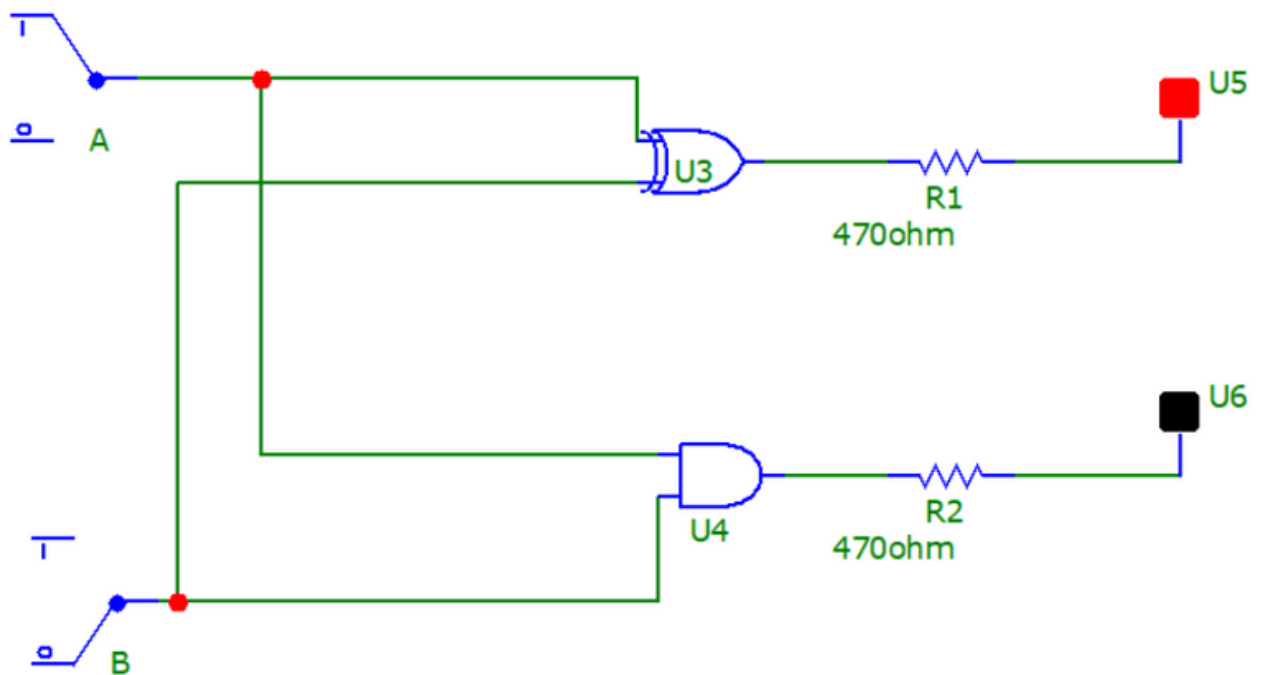
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1. Simplify the sum of product functions for the two outputs

$$C = A.B$$

$$S = A \oplus B$$

2. Draw a logic diagram and Implement a half-adder circuit.



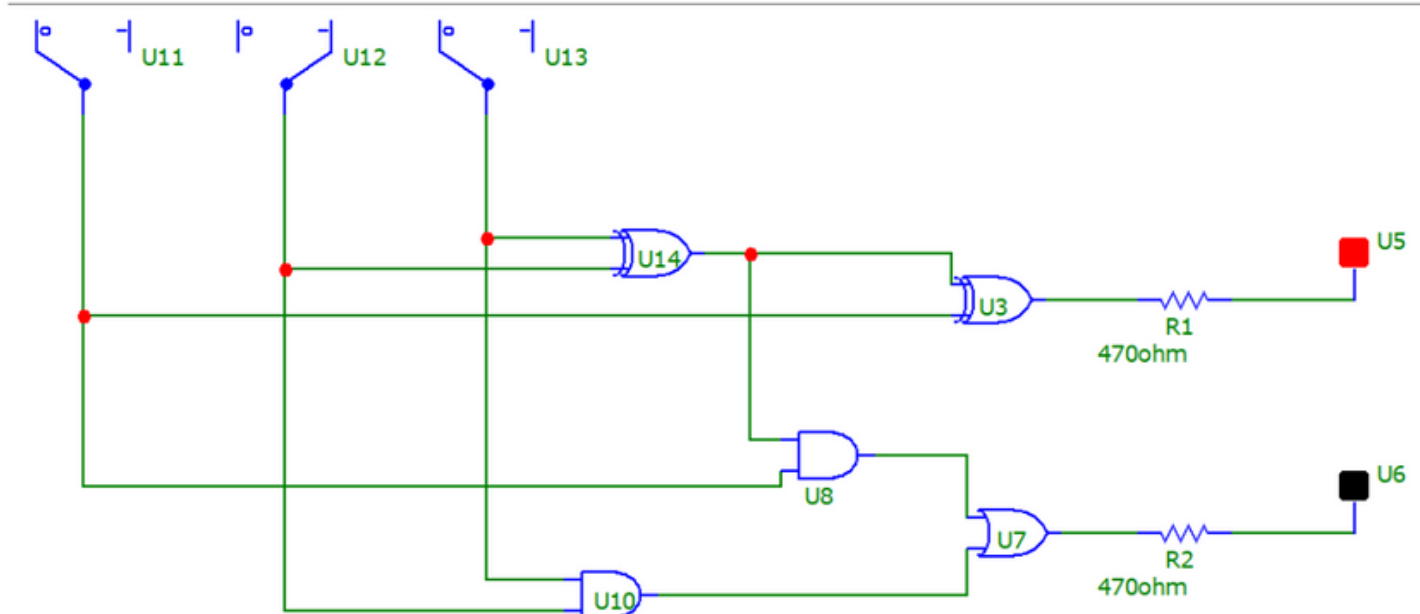
# Part B

1. Simplify the sum of product functions for the two outputs.

$$+C_{out} = AB + Cin(A \oplus B)$$

$$S = A \oplus B \oplus C$$

2. Draw a logic diagram and Implement a full adder circuit.



# Part C

